

Challenge:	Fragments of Truth
Category:	Audio Forensics / Steganography
Difficulty:	Medium
Tools Used:	Audacity, MorseCode.world, SSTV-Decoder, Phone/Online QR Scanner/Google lens

Summary

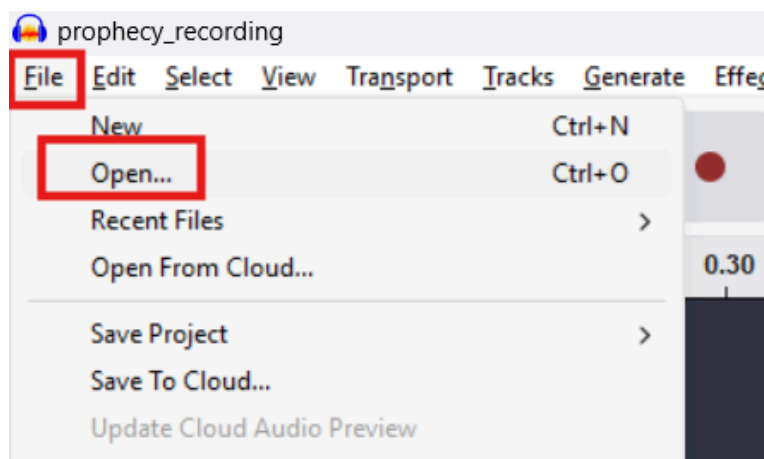
This challenge gives you a single audio file that sounds like complete chaos - static, random beeps, and a mechanical screech. But that chaos is intentional. Hidden inside are three completely different encoding techniques stacked on top of each other: a visual message embedded in the sound frequencies, a Morse code message hidden in the beeps, and an image transmitted through audio tones. Each layer gives you a clue that leads to the next, and the final layer reveals the flag.

Problem Statement

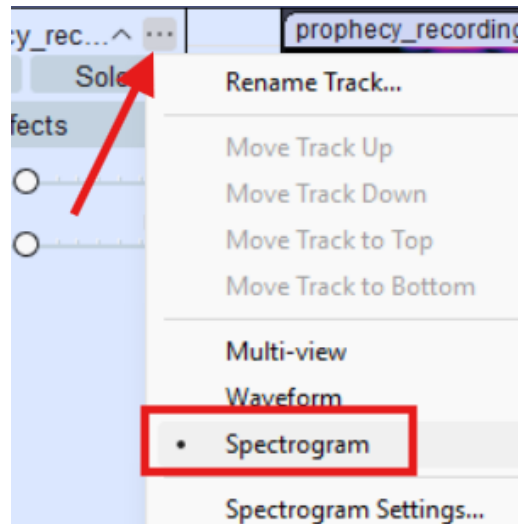
Deep in the Department of Mysteries, SNAPES left behind a prophecy recording. It was sealed in a time-locked vault, meant to be opened only by those with the skills to understand its layers. The audio sounds like chaos - static, beeps, strange tones. But within the noise are three hidden fragments of truth. Decode all three to unlock the FIENDFYRE fragment.

Steps to Solve

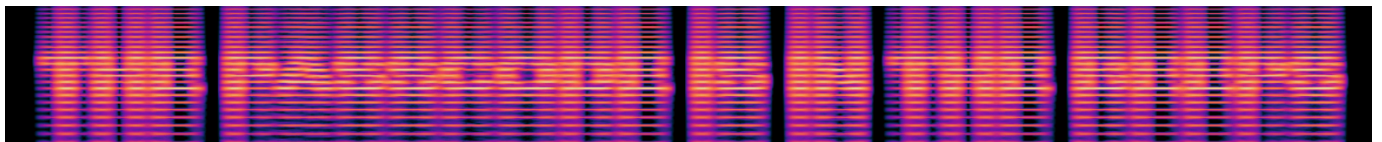
1. Download the audio file from the drive link provided in the challenge.
2. The audio file only produces static when played, so nothing useful can be extracted by listening alone.
Hint 1 Connection: The hint "the eyes perceive what the ears cannot" directly suggests that the solution is visual not auditory.
This indicates that the data is hidden in a visual representation of the audio.
3. So now open Audacity
4. Go to File → Open and load prophecy_recording.wav



5. Click the three dots on the left side of the track
6. Select Spectrogram



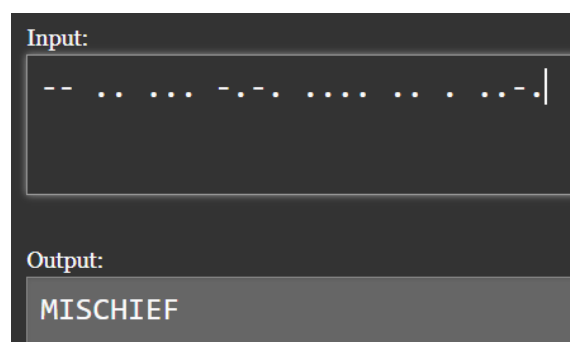
7. Zoom in vertically using Ctrl + Scroll
8. Adjust gain/window size in spectrogram settings if the text isn't clear
9. You will see text appear visually within the frequency bands of the audio:
THE PASSCODE IS IN THE BEEPS



10. This tells you the next layer is hidden in the beeping sounds within the audio.
11. Now that you know the passcode is in the beeps, listen carefully to the middle section of the audio. You'll hear a series of short and long tones — this is Morse code.
12. Note down the pattern of dots (short) and dashes (long):

-- -.-.-.

13. Decode each group using the Morse code manually or using online decoder:
Used morsecode.world



Hint 2 connection: "Short and long breaths — an old Muggle tongue born of wire and telegraph." — Morse code was invented for telegraph communication, exactly as the hint described. Short = dot, long = dash.

14. The final section of the audio sounds like a harsh mechanical screech, completely different from both the static and the beeps. This is SSTV (Slow Scan Television), a technique used to transmit images over audio by encoding pixel data as tones.

15. So to decode it use an online SSTV decoder
16. The decoder scans the audio line by line and reconstructs an image
Used sstv-decoder.mathieurenaud.fr/ here



Hint 3 connection: "The voice of a dying machine... a phosphor screen... line by line."
- SSTV literally draws images line by line using audio tones, originally displayed on phosphor CRT screens.

17. The image produced by the SSTV decoder is QR code, scan it using your phone camera or an online QR scanner or scan it using google lens
18. Save the image from the SSTV decoder
19. Upload it to the QR scanner above or point your phone camera at it
20. The QR code reveals the final flag

Hint 4 connection: "A labyrinth of black and white pixels... only an observing lens of the modern age." - a QR code is literally a matrix of black and white pixels, and your phone camera or online scanner is the modern observing lens.

21. Final Flag:
CRUXHUNT{pr0ph3cy_d3c0d3d_thr33_l4y3rs_r3v34l3d}